

COURSE

OVERVIEW

TOWER ERECTOR

The Tower Erector course is designed to provide learners with the knowledge and practical skills required to safely erect, modify, and dismantle telecommunication towers and tower structures.

The course focuses on structural components, sequencing of erection activities, safe climbing and positioning, fall protection systems, and teamwork during tower construction. Emphasis is placed on hazard identification, safe work methods, correct equipment use, and coordination between team members to reduce the risk of falls, structural failure, and injury during tower erection activities.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Identify hazards associated with tower erection activities
- Apply safe climbing, positioning, and work-at-height techniques
- Select, inspect, and use fall protection and climbing equipment correctly
- Follow correct erection and dismantling sequences
- Work effectively as part of a tower erection team
- Apply safe work procedures during tower construction operations

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GRAVITY

VALUES



R

RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.



I

INTEGRITY

Conducting day-to-day business with integrity at all times.



S

SOLUTIONS **DRIVEN**

Being solution driven.



Q

QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

Before any work begins, the site must be safe.

Use this simple checklist

Follow the Rules:

Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

01



02



Rescue Plan

Have a rescue plan ready. A rescue kit and first aid kit must be on site.

Work in Pairs

There must be at least two workers on site.
If something goes wrong, one can help or call for help.

03



05

Safe Site

Check the building or structure.

Make sure it is strong and safe to work on.



Check Equipment

All tools and gear must be safe and working before use.

04



06 Manage Traffic

If there are cars or trucks nearby, there must be a traffic safety plan.

06



07 Do a Risk Assessment

Check what dangers there are.

All workers must understand the risks before starting work.

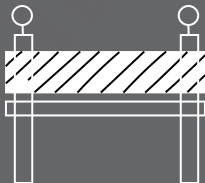
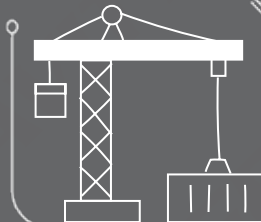
07



08

08 Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES & BARRICADES

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY



Be healthy and ready before starting work.



Only do jobs you are trained and allowed to do.



Never work under the influence of alcohol or drugs.



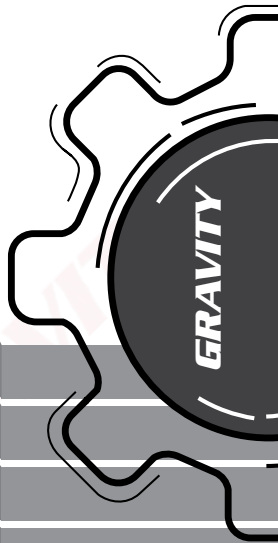
Always wear your safety gear (PPE, harness if working at height).



Use harnesses correctly and stay connected at heights.



Never work alone and always follow safety steps.



SITE SAFETY

Keep people away from areas below when working above.



Don't walk under lifted loads or stand in dangerous areas.



Only trained people can do electrical, underground, or confined space work.



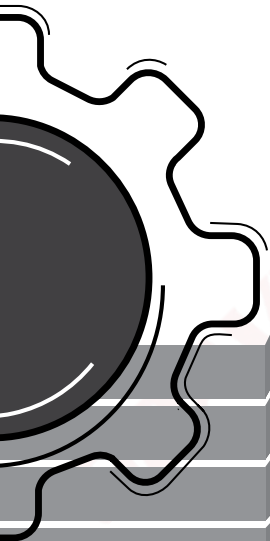
Follow road rules. Wear helmets. No passengers on open trucks..



Don't drive tired or distracted.



Stop unsafe work! Everyone has the right to speak up.





INTRODUCTION TO TOWER ERECTING



This course is designed for people who need to build lattice towers for the telecommunication industry.

***COMPLIANCE WITH
AUDITS AFTER
COMPLETION.***

04

***ENHANCE CLIMBER
SAFETY.***

01

**OUR
GOALS**

***ENSURE HIGH-
QUALITY BUILDS.***

03

***IMPROVE THE EFFICIENCY
OF TOWER CONSTRUCTION
METHODS.***

02

This course does not cover the building of guyed masts, monopoles, or any other similar structures. Although the principles of building a lattice tower stay much the same, specific interpretations and variations to technical drawings should still be considered and engineer requirements will still be applicable to each site. After completion of this course, tower erectors are still required to work under the supervision of experienced personnel who have the competency to erect the required towers.

Upon completion of this course, you will be able to read and interpret tower drawings, understand the components thereof and how to assemble them, and be able to install an entire lattice tower. You will combine the skills of working at height, rigging, theoretical understanding, and physical installation of the tower to achieve this goal.

Why should one do the Tower Erector course.

Tower erection training is essential for several reasons, and the images shared of the skew or incomplete towers highlight some of the potential risks and consequences that can occur without proper training. Towers need to be erected correctly to ensure their structural integrity.

Crooked tower spine



Incorrect hold down bolt installation

In the event of having damaged hold down bolts you cannot level the stubs.

Insufficient bolt protrusion



Excessive bolt protrusion



Misalignment of bolts



Wrong washer size



Excessive stand-off max 5 cm



Damaged threads which will result in the bolts not being able to be threaded on.



Incorrect bolt usage

(Must be at least 8.8 grade bolts)



Missing bolts



Loose components



Missing bracings



Missing planned bracings



Installation alterations not allowed



End results of all of the above-mentioned is catastrophic failure.



In conclusion, tower erection training is vital for safety, structural integrity, efficiency, compliance with regulations, and customer satisfaction.

By providing workers with the necessary knowledge and skills, companies can minimize risks, prevent accidents, and ensure that towers are erected correctly, meeting the highest quality standards

A. Different types of towers.

A: Freestanding lattice towers

A tower that consists of vertical, horizontal, and diagonal members. This could either be CHS or angle iron towers or a combination thereof. Freestanding, also often referred to as self-supporting towers, refers to the fact that the tower is not supported with any cables, it stands on the foundation to which it is anchored.

This course will cover the building of these towers.



B: Monopoles

These are also freestanding towers, however, it is a single pole. A monopole could be accessed via a ladder on the outside, or an internal ladder. They often have crows nests as a platform to work from.



C: Guyed masts: Tall slender structures held up by sets of guyed wires.

B. Tower components

1 : Circular Hollow Sections (CHS):



Round steel tubing that is hollow on the inside.

2 : Angle Iron:



Long steel member usually bent to 90 degrees. On its length, however, it might be bent to 60 degrees.

3 : Flanges:



Steel plates welded to CHS, used to connect two CHS sections together. Flanges can come in different shapes and sizes.

4 : HD bolts (Hold Down bolts):



The bolts that protrude from the foundation, the Stubs connect to these bolts which hold the tower to the ground.

5 : Stubs:



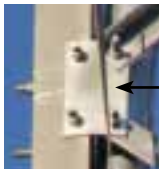
A stub is the first connection from the HD bolts on the foundation to the tower. This connects the tower to the foundation.

6 : Connector plates:



Components that join one leg to another leg.

7 : Splice plates:



These are components used to join horizontal, diagonal members together with a tower leg.

8 : Diagonals:



Members of the tower that joins one leg to another leg, this provides structural strength to the tower and is diagonally positioned.

9 : Horizontals:



Members of the tower that joins one leg to another leg, this provides structural strength to the tower and is horizontally positioned.

10: Legs:



The main structural components of a tower. They are placed vertically, sometimes at an angle, and stacked on top of each other to provide the required height.

11 : Fixed ladders:



A fixed ladder is a type of ladder that is permanently attached to a structure such as a building, tower, or industrial equipment. Unlike portable or extension ladders, fixed ladders are designed to remain in a fixed position and provide a means of access to elevated areas.

12 : Platform:



A tower platform generally refers to a raised structure or platform located on top of a tower. Tower platforms serve various purposes depending on the type of tower and its intended function. Communication towers, such as those used for radio, television, or cellular networks, often have platforms at the top. These platforms allow technicians to access the tower's equipment for installation, maintenance, and repair purposes.

12 : Bolts:



Bolts are mechanical fasteners typically used to join two or more components together. Bolts are commonly made of steel, but they can also be manufactured from other materials such as stainless steel, brass, or titanium, depending on the desired strength, corrosion resistance, or environmental conditions.

13 : Nuts

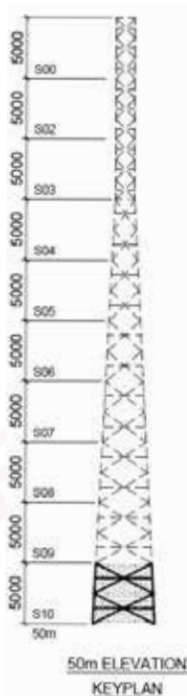


Nuts are mechanical fasteners that are used in conjunction with bolts or threaded rods to secure components together. Nuts are designed to mate with the threaded portion of a bolt or a threaded rod. There are different types of nuts, including standard hex nuts, lock nuts, wing nuts, and cap nuts, among others.

14 : Washers:



Washers are thin, flat, and typically circular discs with a hole in the center. They are used in conjunction with nuts and bolts to provide several important functions in fastening applications.



C. Technical drawings (TD)

Overview

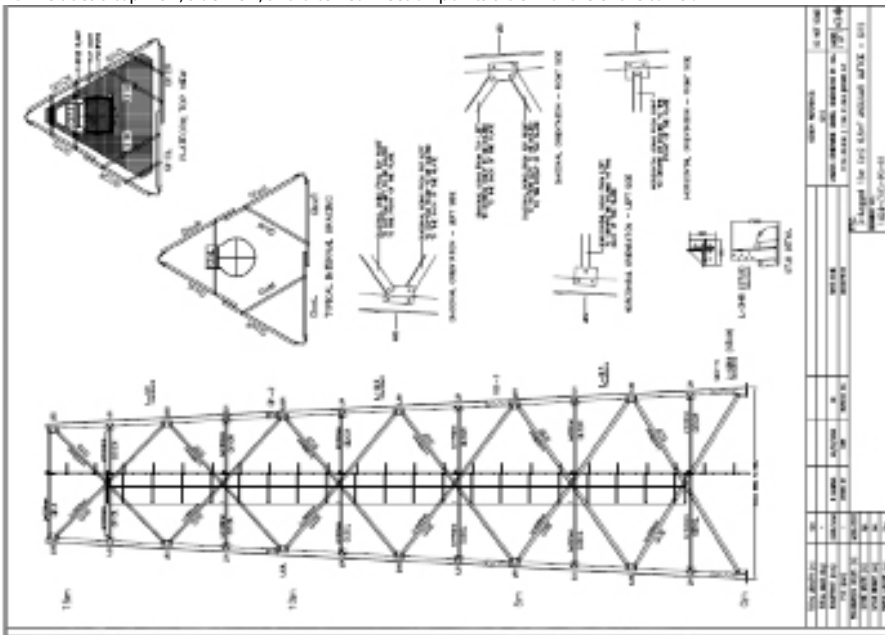
Any tower construction process starts with the (TD). These drawings will give the relevant information regarding:

- Tower profile (type and shape)
- Tower dimensions (height, width etc.)
- Number of sections
- Bill of materials (BOM)
- Material details (weight, dimensions, connection requirements)
- Section details

On the next page is a quick overview of what these main components may look like on technical drawings. Full size drawings will be provided to accompany this manual.

1. General assembly drawing (GAD)

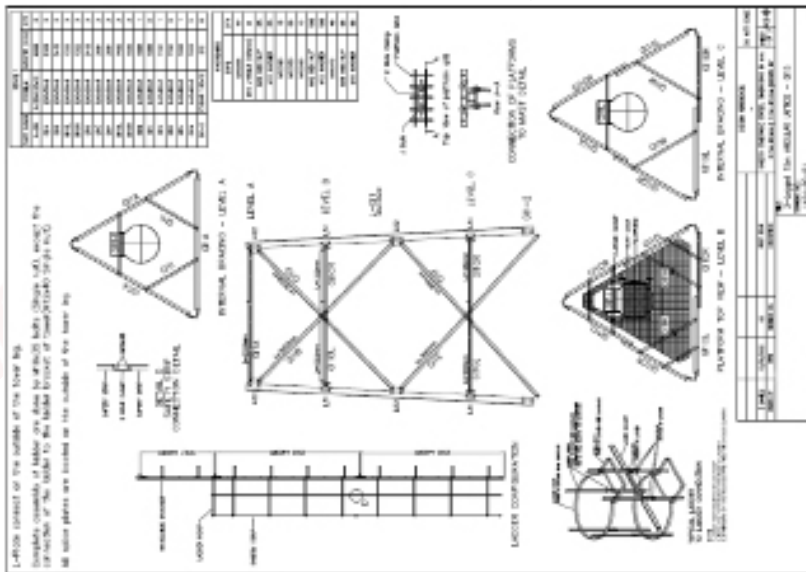
GAD Includes a top view, side view, and often connection points & BOM of the entire tower.



2. Sectional drawing

The sectional drawing is a detailed drawing of a specific section of the tower and what component will be needed to complete the assembly of the section. The sectional drawing allows you to see what each configuration looks like.

It also contains the BOM for that section & often includes the blot assembly and the torque specifications.

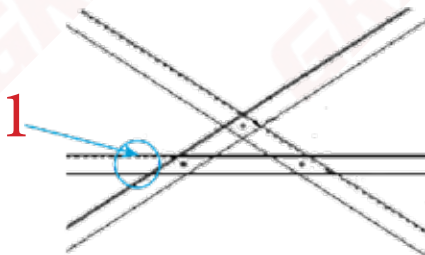


3. Reading drawings

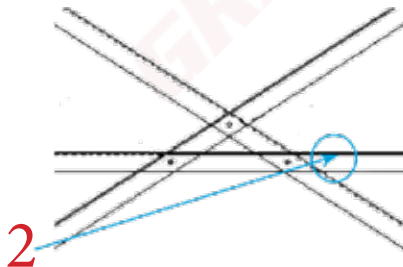
Reading telecoms tower drawings requires a basic understanding of telecom tower design and the associated symbols and conventions used in the drawings. Here's a general guide on how to interpret telecoms tower drawings.

- **Title Block:** The title block provides important information about the drawing, such as the project name, drawing number, revision number, date, and the name of the design firm or engineer.
- **Legends and Symbols:** Telecoms tower drawings use various symbols and abbreviations to represent different components and features. The legend or key typically included in the drawing provides an explanation of these symbols, ensuring you can correctly interpret the drawing.
- **Tower Elevation:** The elevation drawing shows a side view of the tower, illustrating its height, dimensions, and overall appearance.
- **Foundation Details:** This drawing focuses on the tower's foundation, including the type of foundation, dimensions, reinforcement details, and any necessary excavation or site preparation requirements.
- **Structural Framework:** The structural framework drawing displays the tower's support structure, such as the lattice design. It includes details of the steel members, bracing patterns, connection details, and any other structural elements that contribute to the tower's stability and load-bearing capacity.

The dotted line below the small solid line means that the angle iron in number 1 is facing away from you.



The big solid line means that the angle iron in number 2 is facing towards you.



4. Bill of materials (BOM)

The bill of materials (example is given to the right) gives information regarding the components required to construct a given section. This gives critical information such as:

- Component description
- Quantity
- Dimensions (profile)
- Length
- Weight
- Type / Material

The bill of materials also gives the total weight of the section as well the total weight of all the connecting parts (bolts, washers etc.) and the grand total weight of all the components and connection parts.

Note that a section (in the example to the right the section is S10) may be located in the middle of a tower or be the base of a tower. Should it form the base the bill of materials gives the extra components needed to construct the base, such as extra legs, base plates etc.

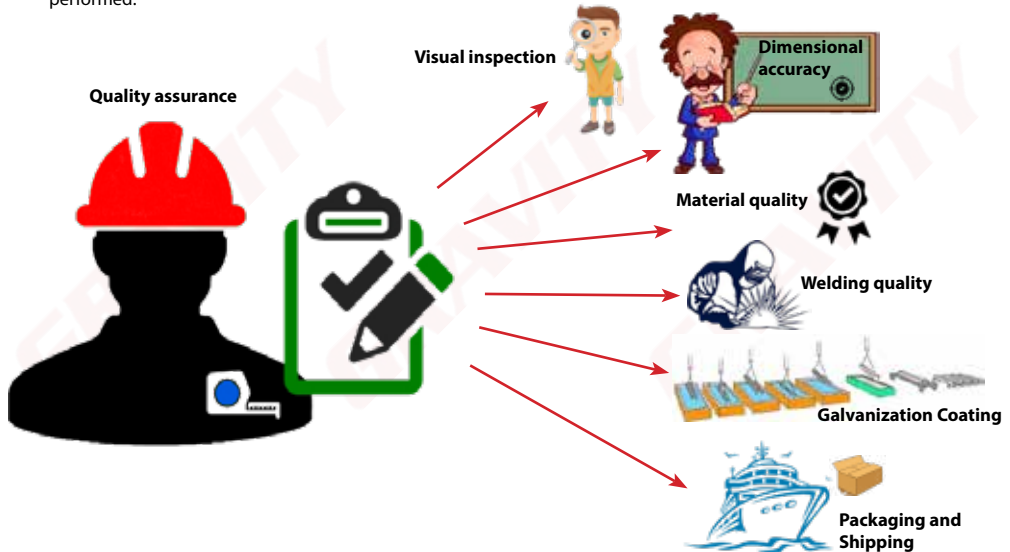
The starting point of any tower construction relies on the bill of materials which is included in the technical drawings, as discussed in the previous section. This must be used to identify and select the correct type and quantity of materials.

Once the correct materials for the stage have been selected, the stage should be laid out in a logical order to facilitate an efficient construction process. This process should not be neglected or disregarded. Proper planning prevents poor performance! It is critical to make sure all components of the tower are in fact present and laid out correctly.

BILL OF MATERIALS					
S10					
Pos No.	Description	Qty.	Profile	Lg(mm)	WT.
1101	Leg	3	L160*160*14	5000	516.6
1102	Cover plate	12	PLT12*120	538	72.98
1103	Plate	3	PLT8*179	203	6.85
1104	Plate	3	PLT8*179	203	6.85
1105	Plate	3	PLT8*180	198	6.71
1106	Plate	3	PLT8*180	198	6.71
1107	Plate	12	PLT6*64	191	6.93
2101	Diagonal	6	L70*70*5	5287	168.1
2102	Diagonal	6	L70*70*5	5074	161.3
3101	Horizontal	6	L50*50*4	2503	45.06
3102	Horizontal	6	L50*50*4	2371	42.68
TOTAL BLACK WT.				=	1040.89Kg
M16-50	BOLT 8.8	84	M16	50	8.65
M16-60	BOLT 8.8	54	M16	60	6.26
M16-80	BOLT 8.8	72	M16	80	10.37
M16 NUT	Class 8.0	420	N16	-	13.99
W16	WASHER	210	W16	-	3.09
6mm Thk. Pack Washers	24	PW16	-	-	0.86
8mm Thk. Pack Washers	6	PW16	-	-	0.29
TOTAL BOLTS WT.				=	43.5 Kg
GRAND TOTAL WT.				=	1084.4 Kg
50m TOWER BASE ASSEMBLY					
4101	Base plate	3	PLT35*100	400	103.58
4102	Leg	3	L180*180*14	608	62.42
4103	Plate	6	PLT10*145	198	11.39
4104	Plate	3	PLT10*130	196	3.41
TOTAL BLACK WT.				=	180.80 Kg
M16-50	BOLT 8.8	12	M16	50	1.24
M16 NUT	Class 8.0	24	N16	-	0.80
W16	WASHER	12	W16	-	0.18
TOTAL BOLTS WT.				=	2.2 Kg
GRAND TOTAL WT.				=	183.00Kg

D. Quality assurance from the manufacturer.

When telecoms towers are manufactured and prepared to leave the manufacturing plant, several quality checks are typically conducted to ensure that the components meet the required standards and correct quantities are supplied. The specific checklist may vary depending on the manufacturer and country, but here are some common quality checks that are typically performed:



Quality assurance steps

1. **Visual Inspection:** The components are visually inspected to identify any visible defects or damages, such as scratches, dents, or misalignment.
2. **Dimensional Accuracy:** The measurements of various components, such as tower sections, mounting brackets, and fasteners, are checked to ensure they meet the specified dimensions.
3. **Material Quality:** The quality of the materials used in the manufacturing process, such as steel or aluminium, is checked to ensure they meet the required standards and specifications.
4. **Welding Integrity:** If welding is involved in the construction of the tower, the quality of the welds is inspected to ensure they are strong, free from defects, and meet the required standards.
5. **Galvanization Coating:** In many cases, telecoms towers are galvanized to provide corrosion resistance. The galvanized coating thickness is measured to verify that it meets the specified standards.
7. **Painting and Finishing:** If the towers are painted or have any other finishing applied, the quality of the paint job, surface finish, and overall appearance is checked for consistency and adherence to specifications.
8. **Documentation Review:** All relevant documentation, such as manufacturing records, test reports, and certifications, is reviewed to ensure that all required processes and quality standards have been followed during the manufacturing process.
9. **Packaging and Shipping:** The components are inspected to ensure they are properly packaged and protected for transportation, following the recommended guidelines to prevent any damage during transit.

Notes: _____

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E. Site establishment.

Delivery of components on-site

All components need to be inspected and verified that the correct parts have been delivered and checked to be in adequate condition.

Check bill of quantities and spot check using a tape measure and vernier.

Check lists

Section No: 38060-S-01			Drawing No: 38060-S-01				Revision No. / Date: 2020-03-28	
Sr. No	Parameter	Specification		Inspection Method	Observation			Remarks
		Nominal	Tolerance		A	B	C	
1	Leg Angle Size	180x80x5	±1	Vernier Caliper	180.1x80.0x5.0	179.9x79.9x5.0	179.9x80.0x5.0	✓
2	Leg Angle Length	4000	±2	Measuring Tape	4001	4001	4001	✓
3	Leg Hole Dia	Ø18	0.0/+0.5	Vernier Caliper	Ø17.8	Ø17.86	Ø18.13	✓
4	Diagonal Angle Size	150x50x5	±1	Vernier Caliper	150.1x50.2x5.1	149.9x50.2x5.1	150.0x49.9x5.0	✓
5	Horizontal Angle Size	150x50x5	±1	Vernier Caliper	149.5x49.8x5.0	149.7x50.1x5.0	150.2x49.9x5.1	✓
6	Top Horizontal Angle Size	150x50x5	±1	Vernier Caliper	149.7x49.7x5.0	149.9x50.2x5.1	150.0x49.9x5.0	✓
Elevation								
7	Diagonal Angle Hole to Hole (2011)	3002	±1	Measuring Tape	3002	3001	3001	✓
8	Diagonal Angle Hole to Hole (2012)	3011	±1	Measuring Tape	3012	3013	3011	✓
9	Horizontal Angle Hole to Hole (2011)	972	±1	Measuring Tape	972	973	972	✓
10	Top horizontal hole to hole (3012)	1062	±1	Measuring Tape	1062	1063	1062	✓
11	Section width on Top (Back to back)	2008	±3	Measuring Tape	2009	2009	2009	✓
11	All Joints gap	No Gap		Visual	OK			✓
Fasteners								
12	Leg-Leg joints	M16x90		Vernier Caliper	OK			✓
13	Bracing & Horizontal joints	M16x60, M16x70		Vernier Caliper	OK			✓
14	All Fasteners Grade	S.S		Visual	OK			✓

Section No: 3NG60-X-60M				Drawing No.: 3NG60-X-60M		Revision No. / Date: 2020-05-19	
Sr. No	Parameter	Specification		Inspection Method	Observation	Acceptable (Y / N)	Remarks
		Nominal	Tolerance				
1	Ladder						
1.1	Alignment	Should be straight		Visual	OK	Y	
1.2	Connections			Visual	OK	Y	
2	Back Guards						
2.1	Alignment	Should be straight		Visual	OK	Y	
2.2	Connection	Ok / Not Ok		Vernier Caliper	Ok	Y	
3	Rest & Work Platform						
3.1	Nourting	Fitment as per drg.		Visual	OK	Y	
3.2	Door Opening	> 120°		Visual	Ok	Y	
3.3	Handrail Assembly	Fitment as per drg.		Visual	Ok	Y	
4	Work Platform						
4.1	Nourting	Fitment as per drg.		Visual	Ok	Y	
4.2	Handrail Assembly	Fitment as per drg.		Visual	Ok	Y	
5	GSM Mount Assembly						
5.1	Channel Assembly with Leg	Fitment as per drg.		Visual	Ok	Y	
5.2	Channel-1 to Channel-2 Dimension	1294 mm		Measuring Tape	1295 mm	Y	
5.3	Tube Dia & Length	Ø76.1x2000		Vernier Caliper/MT	Ø76.2x2002	Y	
5.4	Tube assembly by U-Bolt	Fitment as per drg.		Visual	OK	Y	
6	MW Mount Assembly						
6.1	Channel Assembly with Leg	Fitment as per drg.		Visual	Ok	Y	
6.2	Channel-1 to Channel-2 Dimension	1500 mm		Measuring Tape	1501	Y	
6.3	Tube Dia & Length	Ø114.3x2000		Vernier Caliper/MT	Ø114.5x2001	Y	
6.4	Tube assembly by U-Bolt	Fitment as per drg.		Visual		Y	

Packing





Count & measure



Check the condition of the parts such as rust and corrosion and for any deformation of the components.

Inadequate welding



Rusted bolts, nuts and washers



**Rust
Corrosion
Deformation**



NB : If any component is rusted, it will not be used.

Before work starts

1. Establish a safe and efficient route to the construction area.
2. Lay out members and components in a logical sequential order to make the actual construction more efficient, count all members as well as all the bolts and nuts to ensure that the assembly is not delayed due to missing items.
3. Use the TD to help with the layout piece by piece per section.
4. All the bracing members and connector plate holes must be checked and cleaned to ensure that it does not have any obstruction for the bolts to pass through.
5. The holes must be sprayed with cold galvanizing spray to ensure that the members do not rust.
6. Ensure that the risk assessment and toolbox talk is done and communicated to all on site.
7. Ensure anchor points are suitable and safe for the tower design.
8. Ensure that the site and engineering drawings are compatible.
9. Ensure that all equipment is inspected before use and safe for use and fit for purpose.
10. Ensure that a rescue plan is in place and a sufficient rescue kit is on site with a first aid kit.
11. Ensure that the painting of the tower is done before erection starts.



**CONSTRUCTION
AREA**

Painting process.

When painting the tower leg/angle iron always make sure not to paint where the tower leg meets with the connector plate, as this will prevent the tower from earthing properly.

NB: The same applies to the where the connector plate meets with the stub and where the stub meets with the earthing connection plate.

F. Bolt sizes, types, and how to use a Vernier Caliper

The TD will specify which type and grade of bolts and nuts need to be used on the tower. One cannot just use any type. The grade of the bolts will be visible on the bolt head. If the wrong bolts are used, it can result in catastrophic failure of the tower.

The size of the bolts can be identified through the use of a Vernier caliper. When measuring the thickness, it is important to measure the bolt on the outside of the thread. When measuring the length, one needs to measure from the inside of the head to the end of the bolt.

Take care to not bend the bolts or damage the thread. One needs to install the nuts correctly on the bolts and use the correct washers according to TD requirements. **(See section for unsafe anchors.)**



NOMINAL SIZE	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD	CE	CF	CG	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR	CS	CT	CU	CV	CW	CX	CY	CZ	DA	DB	DC	DD	DE	DF	DG	DH	DI	DJ	DK	DL	DM	DN	DO	DP	DP	DQ	DR	DS	DT	DU	DV	DW	DX	DY	DZ	EA	EB	EC	ED	EE	EF	EG	EH	EI	EJ	EK	EL	EM	EN	EO	EP	EP	EQ	ER	ES	ET	EU	EV	EW	EX	EY	EZ	FA	FB	FC	FD	FE	FF	FG	FH	FI	FJ	FK	FL	FM	FN	FO	FP	FP	FQ	FR	FS	FT	FU	FV	FW	FX	FY	FZ	GA	GB	GC	GD	GE	GF	GG	GH	GI	GJ	GK	GL	GM	GN	GO	GP	GP	GQ	GR	GS	GT	GU	GV	GW	GX	GY	GZ	HA	HB	HC	HD	HE	HF	HG	HH	HI	HJ	HK	HL	HM	HN	HO	HP	HP	HQ	HR	HS	HT	HU	HV	HW	HX	HY	HZ	IA	IB	IC	ID	IE	IF	IG	IH	II	IJ	IK	IL	IM	IN	IO	IP	IP	IQ	IR	IS	IT	IU	IV	IW	IX	IY	IZ	JA	JB	JC	JD	JE	JF	JG	JH	JI	IJ	JK	JK	KL	KM	KN	KO	KP	KP	KQ	KR	KS	KT	KU	KV	KW	KX	KY	KZ	LA	LB	LC	LD	LE	LF	LG	LH	LI	LJ	LK	LM	LN	LO	LP	LP	LQ	LR	LS	LT	LU	LV	LW	LX	LY	LZ	MA	MB	MC	MD	ME	MF	MG	MH	MI	MJ	MK	ML	MM	MN	MO	MP	MP	MQ	MR	MS	MT	MU	MV	MW	MX	MY	MZ	NA	NB	NC	ND	NE	NF	NG	NH	NI	NJ	NK	NL	NM	NN	NO	NP	NP	NQ	NR	NS	NT	NU	NV	NW	NX	NY	NZ	OA	OB	OC	OD	OE	OF	OG	OH	OI	OJ	OK	OL	OM	ON	OO	OP	OP	OQ	OR	OS	OT	OU	OV	OW	OX	OY	OZ	PA	PB	PC	PD	PE	PF	PG	PH	PI	PJ	PK	PL	PM	PN	PO	PP	PP	PQ	PR	PS	PT	PU	PV	PW	PX	PY	PZ	QA	QB	QC	QD	QE	QF	QG	QH	QI	QJ	QK	QL	QM	QN	QO	QP	QP	QQ	QR	QS	QT	QU	QV	QW	QX	QY	QZ	RA	RB	RC	RD	RE	RF	RG	RH	RI	RJ	RK	RL	RM	RN	RO	RP	RP	RQ	RR	RS	RT	RU	RV	RW	RX	RY	RZ	SA	SB	SC	SD	SE	SF	SG	SH	SI	SJ	SK	SL	SM	SN	SO	SP	SP	SQ	SR	SS	ST	SU	SV	SW	SX	SY	SZ	TA	TB	TC	TD	TE	TF	TG	TH	TI	TJ	TK	TL	TM	TN	TO	TP	TP	TQ	TR	TS	TT	TU	TV	TW	TX	TY	TZ	UA	UB	UC	UD	UE	UF	UG	UH	UI	UJ	UK	UL	UM	UN	UO	UP	UP	UQ	UR	US	UT	UU	UV	UW	UX	UY	UZ	VA	VB	VC	VD	VE	VF	VG	VH	VI	VJ	VK	VL	VM	VN	VO	VP	VP	VQ	VR	VS	VT	VU	VV	VW	VX	VY	VZ	WA	WB	WC	WD	WE	WF	WG	WH	WI	WJ	WK	WL	WM	WN	WO	WP	WP	WQ	WR	WS	WT	WU	WV	WW	WX	WY	WZ	XA	XB	XC	XD	XE	XF	XG	XH	XI	XJ	XK	XL	XM	XN	XO	XP	XP	XQ	XR	XS	XT	XU	XV	XW	XX	XY	XZ	YA	YB	YC	YD	YE	YF	YG	YH	YI	YJ	YK	YL	YM	YN	YO	YP	YP	YQ	YR	YS	YT	YU	YV	YW	YX	YY	YZ	ZA	ZB	ZC	ZD	ZE	ZF	ZG	ZH	ZI	ZJ	ZK	ZL	ZM	ZN	ZO	ZP	ZP	ZQ	ZR	ZS	ZT	ZU	ZV	ZW	ZX	ZY	ZZ
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G. Tower construction

Foundation compliance checklist.

The foundation needs to be inspected before any tower construction can commence. Below is an example of inspection criteria.

Criteria	Yes	No	N/A	Comments
Levelness of concrete columns, plinth, and remaining foundation to not allow for standing water.				
Backfill, compaction, and site dressing were done correctly.				
Signs of erosion around the base of the tower and the foundation.				
Spacing/dimensions.				
(Hold Down) HD bolts – Plumbness/size/correct quantity of nuts				
(Hold Down) HD bolts are torqued according to the correct specifications.				
Correct thickness of non-shrink grouting used (At least 30mm)				
Rust on (Hold Down) HD bolts				
(Hold Down) HD bolt threads protruding (2 threads minimum)				
Repairs are done on non-structural damage to concrete.				
The tower legs are properly bonded to a grounding ring/ grounding grid.				
Grounding connection from the foundation securely fastened to the stubs.				

1. Cast anchors and inspections

Before starting any construction, the anchor points on site must be inspected.
There are two types of anchors for tower building on site:

1. Cast plinth anchor for the hoist.
2. In-situ anchors for tower base.

These anchors must be checked according to the foundational compliance checklist to ensure the anchors are sufficient.

NB: the concrete used in the foundation needs to cure for about 7 days to reach a 30 % strength, 14 days to reach 50 % strength and 21 days to reach 100 % strength.



In-situ anchors for tower

When constructing a tower, it is of utmost importance to ensure that the starter anchors and base are inspected and are in a safe condition.



Inspect foundation for cracks, levelness and size as per engineering drawings.

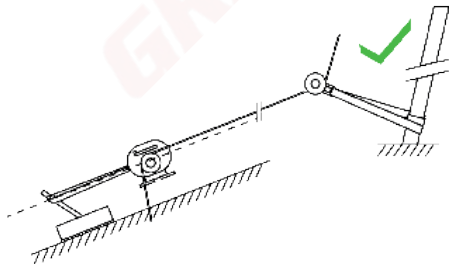
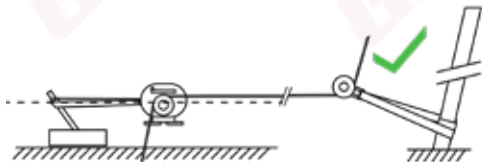
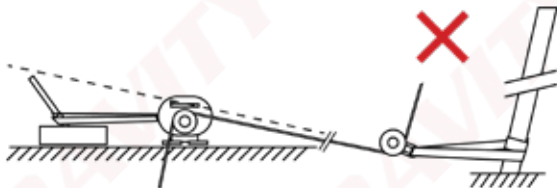


Inspect in-situ cast anchors for plumbness and distances between are as per engineering drawings. Inspect thread for signs of damage.

Cast plinth anchor for the hoist

Because the anchoring points for tower construction are casted plinths to the side of the tower, it is important to ensure the hoist is anchored as to prevent swinging, whilst allowing easy control of the hoist.

1. Ensure that when loaded the load-line is as straight as possible and as in-line as possible with the anchor point. The rope must never pull in a downward direction as the frame of the hoist is not able to handle downward pressure and it might break.
2. When working on sloped surfaces, ensure the load line is still in-line with the anchor sling of the hoist.



Notes: _____

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H. Tower overview

1. Tower inspection overview

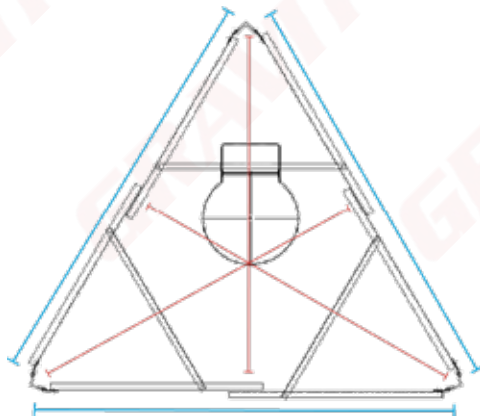
The towers need to be inspected after assembly. It is important to assemble it in such a manner as to comply with the inspection criteria. Below is an example of inspection criteria.

Criteria	Yes	No	N/A	Comments
Applicable tower drawings are available on-site during the inspection.				
All bolts are present and protruding (2 threads minimum)				
All members are marked correctly according to the technical drawing.				
All washers are present and installed correctly.				
All nuts are present and installed correctly.				
All Stubs are present and installed correctly				
All members are present and are the correct type, size, and orientation.				
All connector plates and splice plates are present and are the correct size and orientation.				
All bracing is correctly assembled and installed.				
All tower members are straight.				
The paintwork is adequate (Galvanizing).				
Rust or corrosion was found on the tower.				
No twists or misalignments in the tower.				
Platforms and ladder are installed.				
All stages are square and level.				
All earth cables present and are connected to the stubs.				
Lightning conductor correctly installed.				

2. Section inspection

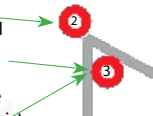
Once a section has been completed it must be inspected to ensure that it is safe and stable in order to install the next section. If the current section is not safe or stable the subsequent sections will also not be safe or stable, also, it might cause the tower or gin pole to fail.

After the second section of the tower is built it needs to be torqued (see section J on page 53-56 for torquing specifications). Build the rest of the tower and hand tighten the bolts as you go up. Once done, torque the rest of the tower starting from the second bottom section, going up

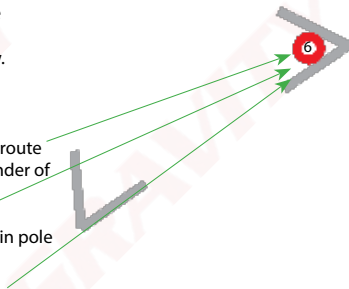


3. Tower erection overview

- Step 1 Install and level the stubs on the hold down bolts, then connect the correct connector plate to each stub.
- Step 2 Install the Gin pole on the stub closest to the hoist plinth and lift the first leg with a **pivot** action.
- Step 3 Move the Gin pole to the inside of the first leg and install the two supporting bracings. One on each side of the first leg and place the Gin pole in a **diagonal orientation** and secure it with a Lever hoist.
- Step 4 Using the Gin pole install the remaining two legs with the same **pivot** action as leg one and install all the remaining bracings.
- Step 5 Measure the first stage of the tower to ensure that it is not skew.



- Step 6 Move the Gin pole to one of the opposite legs to create a clear route for the lifting rope. The Gin pole will remain on this side for the remainder of the build.
- Step 7 Gin pole will be in a vertical orientation. Install the 4th leg.
- Step 8 Jump the Gin pole to the next horizontal level and install the Gin pole in a diagonal orientation for leg five and six.
- Step 9 Repeat step 7 and 8 to complete the tower erection.
- Step 10 Install aircraft warning light, ladders, platforms and lightning conductors.
- Step 11 Torque the tower.



NB. The top of the Gin pole should be high enough to allow enough freeboard to lift the leg, however, if the Gin pole is too high it will place too much force on the Gin pole.

A good indication is that the Gin pole should be the same height as the top of the leg member will be once installed.



4. How to use the Dumpy level.

The dumpy level is an optical surveying leveling instrument and is used to determine relative height and distance among different locations of a surveying land. It is used together with a tripod stand and a measuring staff.

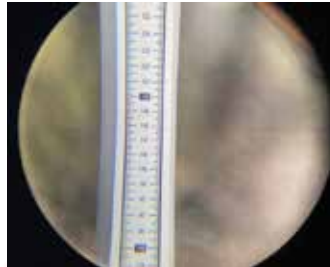


There are a few crucial things that you have to do when using a dumpy level.

1. Rig the tripod by opening the legs and making sure that the top of the tripod is more or less shoulder height.
2. Take the tripod top cover off, ensure that the legs of the tripod are firmly spiked into the ground, and ensure that the top of the tripod is level.
3. Place the dumpy level on the tripod at eye height and fasten it with the bolt on the tripod.
4. Level the dumpy level with the level vial /bubble tube that is on the left side of the dumpy level ensure that the bubble is in the middle of the small circle of the vial.
5. Rotate the dumpy level in 2 other directions to ensure that the bubble remains in the middle of the circle.
6. If you need to adjust the level, you can do this by making small adjustments by turning the leveling screws on the dumpy level. Always make sure that the dumpy level telescope points to a single leveling screw, and the lens side of the dumpy level should be in the middle of the remaining 2 leveling screws when leveling the dumpy level.
7. Once the dumpy level is set up correctly do not touch the dumpy level or the tripod.
8. One person will then take the measuring staff and hold it at the point that needs to be measured with the millimeter side of the staff facing towards the dumpy level. Then ensure that the bottom of the staff is clean to prevent a false reading.



Out of focus



In focus

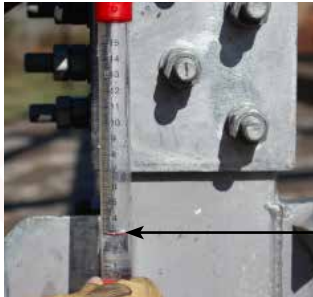


9. Once the measurement has been determined one will look through the lens one more time to confirm the correct measurement and then the staff will be moved to the next point that needs to be measured. Repeat the measuring process.



5. How to use the Egyptian water level.

An Egyptian water level is a simple tool that uses a water-filled tube to determine level or slope. Here are the steps on how to use an Egyptian water level.



There are a few crucial things that you have to do when using a Egyptian water level.

1. Set up the equipment: Gather your Egyptian water level, a measuring tape, two stakes, and a level.
2. Locate the tower stubs: Identify the locations of the tower stubs that need to be leveled.
3. Place one end of the tube on a fixed reference point that you want to work from.
4. Take the other end of the tube to the next point that you want to measure, at this stage ensure that you estimate that both points should be more or less the same height.
5. Ensure that both ends of the tubes are open and facing towards the top. Then pour water into the tube until it is almost full on both ends and ensure that there are no air bubbles in the tube.
6. Take measurements: With the water level attached to the tower stubs, measure the height of the water level at each stake. Record the measurement for each stake.
7. Check the measurements: Use a level to check the height of each secure point. Adjust the height of the tower stubs to match the height of the water level at each secure point.
8. Re-measure and adjust: Repeat steps 6 and 7 until the measurements match and the tower stubs are level.
9. Remove the equipment: Once the tower stubs are level, remove the water level, stakes, and measuring equipment.

6. Stubs and how to install them.

Importance of levelling stubs.

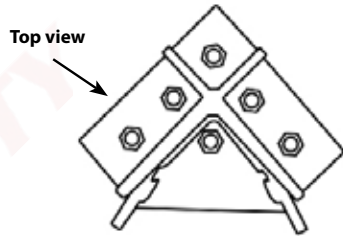
Stubs are the first connections from the HD bolts on the foundation to the tower. This together with the stubs are the brackets that connect the tower to the foundation and the stand-off between the foundation concrete and the stub should not be more than 50 millimeters.

Stubs must be levelled to ensure that the tower is built straight. If stubs are not levelled the tower misalignment grows exponentially as the tower goes higher. This could result in catastrophic failure of the structure.

Level Stubs using a Dumpy Level

There are many ways one can use to ensure that the stubs are level.

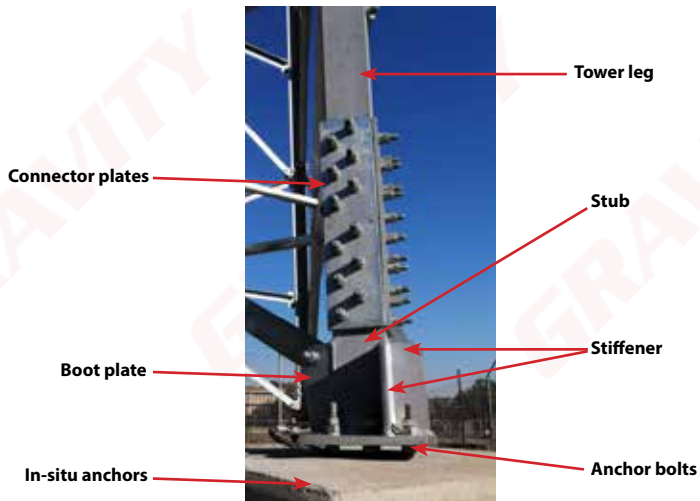
The most common way however is to use a dumpy level. This is a cheap and efficient way to ensure correct levels.



1. Use the dumpy level to determine the highest point in the foundations. This will determine the point from which to level the tower.
2. Then use the dumpy level to ensure all hold down bolts are more or less the same height, they should preferably have no more than 3 mm of tolerance from the determined height.
3. Stubs need to be levelled by using only the outer hold-down bolts on the three corners, by using this method and with the help of a spirit level, it is faster to get the overall level of the stubs. One will have to move the measuring staff to 4 different positions (front, back, left, and right) on the stubs to ensure that the stub is measuring the same height all around.
4. Once the stub is level the top nuts of the hold-down bolt will be tightened, and the middle of each hold-down bolt will be raised until they touch the bottom of the stubs and will be tightened with the spanner. The top nut above the stub on the hold-down bolts will then also be tightened. Repeat the same process for all the stubs and double-check the level for all stubs and ensure that the difference is no greater than 3 millimeters between all the stubs.

7. Assembly of connector plates onto the stubs:

First, one would need to connect the connector plates to the legs. Use the TD to ensure you connect these plates correctly. At this stage, the plates will be relatively loose so as to allow the installation of the rest of the members.



Notes: _____

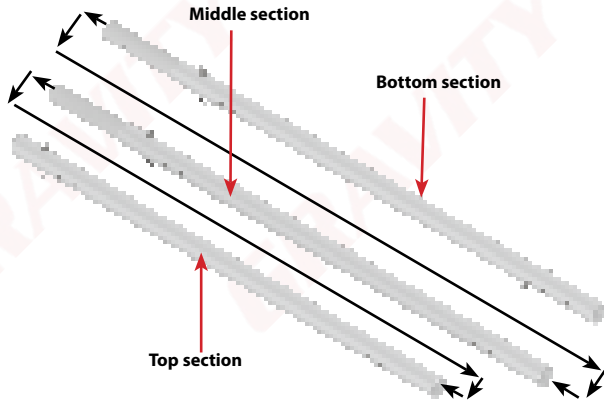
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8. Gin pole assembly

Gin poles are often times delivered in a lot of different pieces and has to be assembled before using it, consult the manufacturers assembly booklet to assemble the gin pole correctly.

When assembling a 9-meter gin pole, it is important to ensure that you correctly assemble the 3 different sections of the gin pole so that the final assembly can be done correctly. This is a crucial stage for the orientation of the eye-nuts and bolts which are different for some sections. If this is not done correctly you will not be able to complete the assembly.

- When assembling the bottom stage, the eye-nuts that are installed at the bottom will be the only eye-nuts installed in this direction (same as all the main bolts that keep the gin pole together).
- The second set of eye-nuts will be installed in a different direction and all other eye-nuts will then be installed in the same direction.
- The installation to the right will show the 3 different sections and the correct orientation/direction that all the eye-nuts and main bolts will be installed.
- Once all sections have been assembled correctly, one will join them together by taking the pre-assembled bottom part and placing it into the bottom of the middle section and allowing the holes to line up. Then join the pre-assembled top section and place it into the



top part of the middle section once again ensuring the holes for the main bolts line up. (Should a Gin pole be longer the only thing that will differ is the second or even a third middle section that will be assembled the same way).

9. Gin pole construction forces

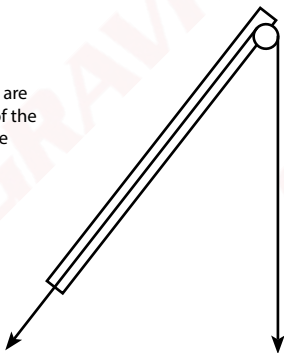
Gin poles are often designed to be used in a vertical configuration, not diagonal or horizontal. Before using it in such a configuration, get manufacturers approval for the forces and angles it can handle.

When using a gin pole it is of utmost importance to be aware of the forces involved and how they are applied. This will prevent the gin pole from being used incorrectly or being used outside of its design limits.

When using a gin pole, it is important to ensure that the gin pole is DIRECTLY above the area where the member is to be fixed. If not the forces on the gin pole may exceed the WLL of the pole, the pole anchors or the tower itself which could lead to disastrous consequences.



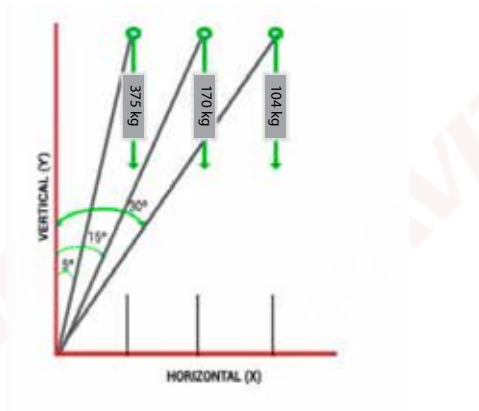
When the gin pole is vertical the forces are distributed through the entire length of the pole, meaning it is much stronger in the vertical orientation.



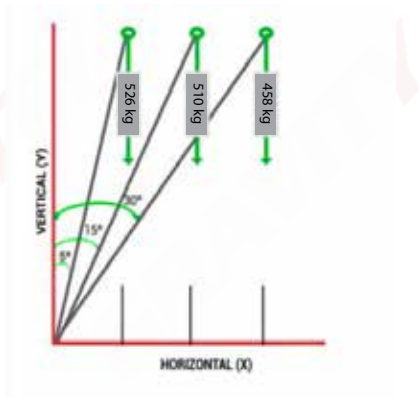
When the gin pole is tilted or angled away from the vertical, the force applied to the tip of the pole and the force experienced at the base increase dramatically as well as increasing the bending within the pole which may cause it to fail.

10. Examples of Gin pole safe load charts

9 m Gin pole load chart



12 m Gin pole load chart



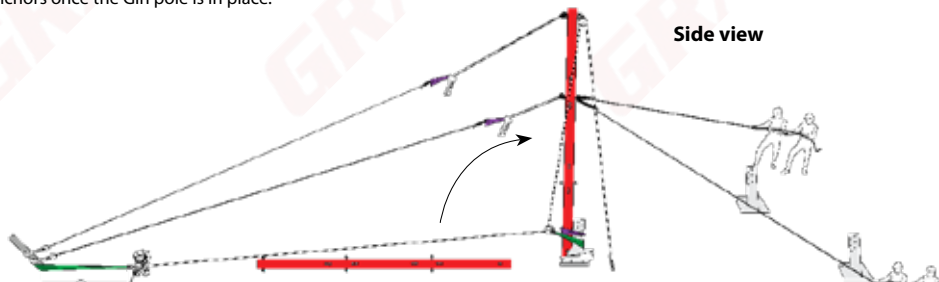
WARNING: Ensure that you obtain the load chart from the original equipment manufacturer and read the instructions before using it, as the WLL may differ between manufacturers.

11. Installing the Gin pole - for first leg

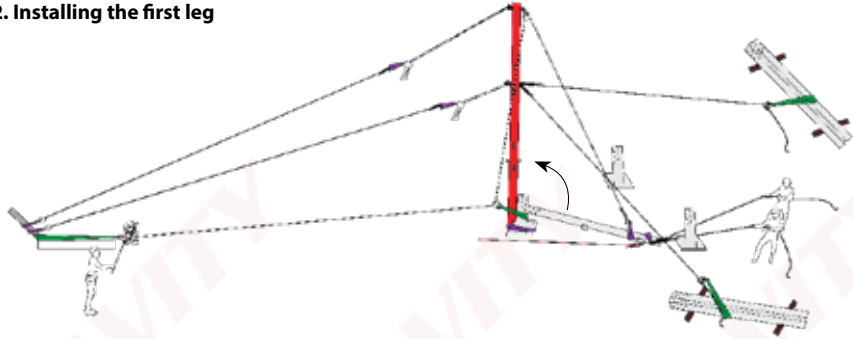
When constructing a tower, the first section is critical as this will influence the stability and strength of the entire tower.

Before erecting the first member and after the installation and leveling of the subs ensure the following.

1. The Gin pole must be installed before the lifting of the first leg can take place.
2. The lever hoist with the three-meter chain and two-meter lifting sling will be connected to the top of the Gin pole on the hoist anchor plinth side.
3. One of the anchor lines will be connected to the lever hoist that is connected to the top of the Gin pole while the second anchor line will be connected to the eye-nut one level lower where the lever hoist is connected. Two guide ropes will be connected to the other side of the Gin pole that is on the same level as the second anchor line that will be used to stabilize the Gin pole, while a lifting rope will also be connected to the Gin pole with two pulleys. One pulley will be connected to the bottom of the Gin pole facing towards the hoist anchor plinth and the other pulley will be connected to the top of the Gin pole that will face the opposite side of the hoist anchor plinth.
4. The bottom of the Gin pole will be secured to the stub closest to the hoist anchor plinth. Ensure the end of the lifting rope is secured to the bottom of the Gin pole to prevent it from running up.
5. Then two-four people will lift the Gin pole and walk it inwards to lift the Gin pole upright while others support the Gin pole with the guide ropes.
6. The remaining two guide ropes will be held by four workers, two on each side and must be secured to the dead-weight anchors once the Gin pole is in place.



12. Installing the first leg



1. Install the vertical lifeline while the member is on the ground and before it is lifted into place.
2. Align the tower leg with the base anchor and install one bolt through the appropriate bolt hole.
3. Install two guide ropes to the tower leg to ensure that it does not twist or bend while lifting.
4. The lifting rope from the Gin pole must be secured to the top of the leg.
5. When lifting the first leg the hoist needs to be secured to the hoist anchoring plinth with the use of a lifting sling and a bow shackle, the lifting rope will be connected to the hoist that will go through the pulleys that are connected to the Gin pole and then connected to the first leg. Lift the leg using the Gin pole.
6. Pivot the tower leg on the one bolt installed on the base anchor, use the guide ropes to ensure that the leg is kept straight, until it is in the correct position and install the remaining bolts. At this stage, only hand tighten the leg bolts.
7. After the first leg is installed, one complete bracing needs to be installed to support the first leg and to support the Gin pole by lifting the other legs into place. **(NB Check that the bracing is the correct stage bracing and in the correct orientation before installing).**
8. Please note, that if you are building a tower with CHS members, the legs will have flanges at the bottom and top of each leg. Thus, the above-mentioned procedure will not be applicable. One would need to either use a tripod (or a crane) to lift the first legs and place it on the hold-down bolts. Take care here to not damage the thread of the HD bolts.

13. Assemble horizontal and diagonal bracing:

Using the TD, one can pre-assemble the diagonal and horizontal members together on the ground. One can also connect these members to the legs of the tower before assembly. It all depends on your mode of operation; however, one needs to keep in consideration the weight of the combined members.



The bracing may be assembled on the ground before lifting it in place, so the correct procedure should be:

1. Identify and select the correct components as per bill of materials.
2. Lay out components in the order they are to be constructed.
3. Assemble the bracing.
4. Install the legs of the tower.
5. Install the bracing.
6. Repeat steps for each section.

Connecting sequence for installing bracing to the tower.

When facing the tower and having the bracing in the correct orientation always line up the bracing with the splice plates in the following order.

1. Connect the top side on the inside of the tower splice plate.
2. Then connect the bottom inside to the splice plate.
3. Connect the middle bracing to the inside of the tower splice plate.
4. Connect the middle to the outside of the splice plate.
5. Connect the bottom side to the outside of the splice plate.
6. Connect the top side to the outside of the splice plate.

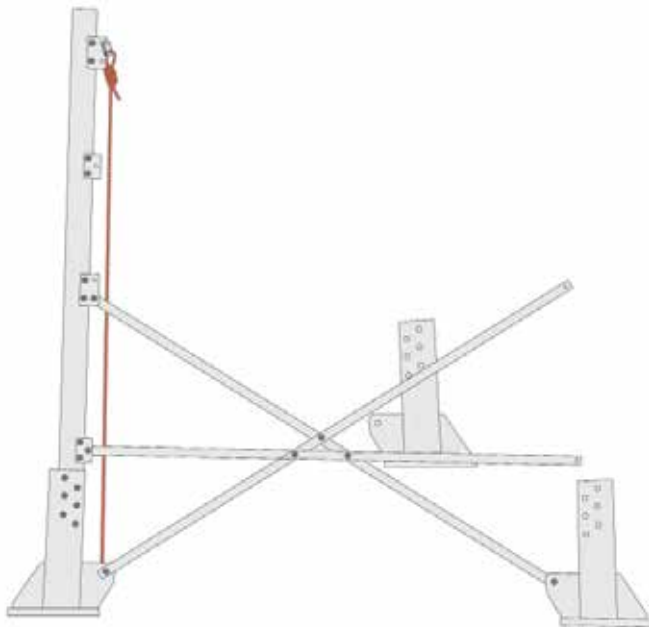
14. Installing the bracing

In order to strengthen and stabilize the tower, bracing needs to be installed on each section before moving on to the next. This does not only fulfill the requirements, but also provides stable footing for the workers to stand on when moving the Gin pole.

1. Install all bracing by lifting the bracing into place and hand tighten the bolts to the first legs. Take care to follow the TD instructions. Once again, only hand tighten the nuts and bolts.
2. Once the first stage leg and bracing have been installed the Gin pole needs to be lowered by having two persons holding the guide ropes while two-four workers will once again have to walk the Gin pole down and move it to the inside of the first leg.
3. Check the stage using dumpy level and tape measure for plumbness and squareness.
Plumbness: The level of each base plate has to be the same elevation, if not, it affects the tower stability.
Squareness: The stubs have to be 100% equal diagonally across the base. If not, the tower is not square.
4. At this stage, you want to ensure that the tower is level using the dumpy level as one can still make minor adjustments. Once the levels are correct, one can tighten the stubs to the first legs by torquing them to the correct torque, however, all other bolts can be hand tighten. Now the HD bolts can be tightened, followed by the torquing of all other member bolts. Consult the TD for torque specifications.



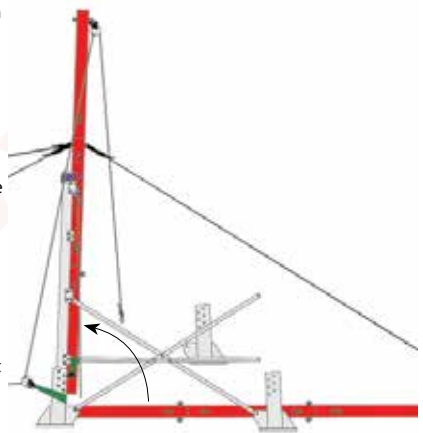
Securing the bracing to support the first leg



15. Securing the Gin pole on the inside of the first leg of the tower.

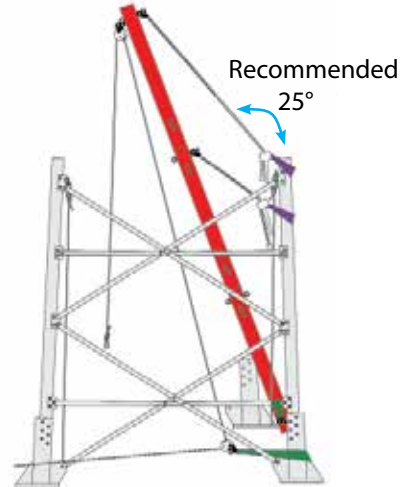
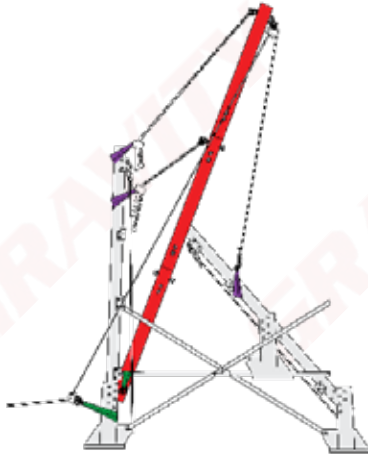
When the first legs are erected and the first level of bracing has been installed, the gin pole needs to be installed on the inside of the first leg in order to facilitate the lifting and installation of the subsequent sections.

1. To secure the Gin pole on the inside of the first leg of the tower only two guide ropes will be secured 2/3 towards the top of the Gin pole. The base of the Gin pole will be placed in the inside corner of the first leg of the tower which will serve as a pivot point for the Gin pole when being lifted in the same way as before and supported with the guide ropes.
2. Once in place, the Gin pole will be secured on the top of the first leg to keep the Gin pole up straight. The second bracing member can then be installed opposite of the first bracing member. By doing this one now has two horizontal members available to secure the Gin pole.
3. The hoist rope will be used to lift the Gin pole slightly and suspend it on the two eye-nuts that are at the bottom of the Gin pole. They will be connected to the slings that are connected on the horizontals of the two bracing members, one on each side.
4. After securing the Gin pole a technician will climb up on the lifeline that is on the first leg and goes up and install two lever hoists that will support the Gin pole when being used in a diagonal orientation. One of the final things that need to be done, is securing the base of the Gin pole to the first leg that will prevent the Gin pole from kicking out and being unbalanced during the lift of the remaining



leg members.

5. The remaining leg members will be lifted in the same way as the first leg by connecting two guide ropes and one lifeline to the top side of the leg member. The bottom of the leg will be secured with one bolt on the stub that will allow the leg to pivot when being lifted and the lifting rope will be secured more or less 2/3 towards the top of the leg member to allow freeboard between the pulley on the Gin pole. Once in place the remaining bolts will be installed, and hand tightens the bolts of the leg members to the stubs.

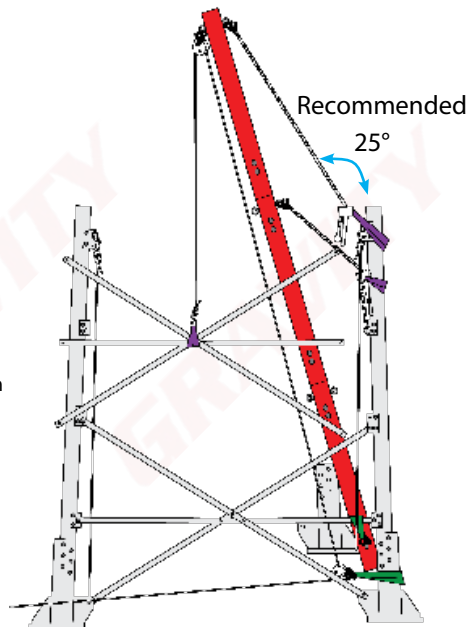


16. Lifting the subsequent members

Once the gin pole is installed the lifting system may be connected, ensure the following:

1. The hoist is securely anchored, and the plinth anchor is safe for use and fit for purpose.
2. The lifting line is deviated on the bottom of the tower.
3. The load is slung safely.
4. The lift route is clear.
5. The top pulley and the load sling position setup provides sufficient space.

Lift the second set of bracing members and secure them according to the TD by making sure that some need to be on the inside of the leg members and others on the outside of the leg members.

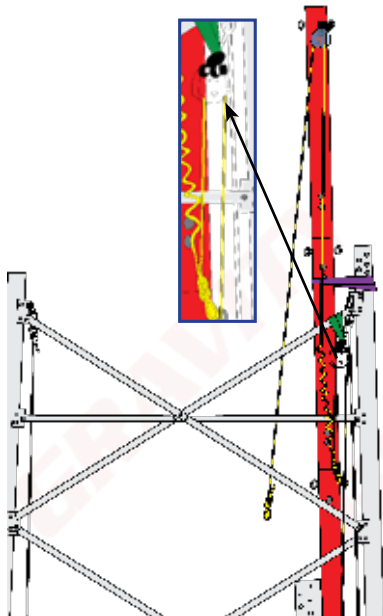


17. Jumping the Gin pole

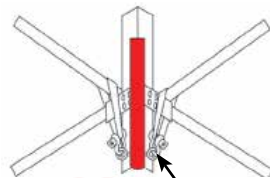
Once the first section has been installed along with the bracing, creating a safe foundation for the rest of the tower, the gin pole needs to be moved up to allow the next section to be installed or “jumped” to the next section.

1. The lever hoist is used to bring the gin pole as close to vertical as possible.
2. A sling is wrapped around the tower leg and gin pole, but still allowing the gin pole to slide up freely.
3. A pulley system is installed at the bottom of the tower.
4. The gin pole is lifted with the pulley system until the next section is reached.
5. The bottom anchor points are connected to two one-ton slings wrapped around the second horizontal bracing of the first stage of the tower and connected to the bottom anchor points of the gin pole with bow shackles at about four meters high for the first jump.

When the next stage needs to be lifted the Gin pole will have to be brought back to its vertical orientation and then be lifted with the same 1:1 pulley system as before to secure the Gin pole on the next horizontal member of the bracing. Once in place, the next stage leg member will be lifted straight up with the necessary guide ropes and lifeline.

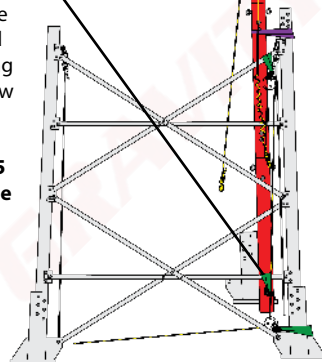


Lifting the gin pole to the next section



Securing the bottom of the Gin pole to two slings and bow shackles while securing the top with a sling and bow shackle.

NB the Gin pole should not stick out more than 5 meters from the top of the supporting sling.



Securing the gin pole to the next section

18. Installing the ladder

There is no specific way one needs to install the ladders. It can be installed as the tower gets erected or it can be installed after the tower is completed. It is however very important to ensure that the ladder gets installed according to the TD requirements and fastened correctly.

19. Tower construction halting

When the tower construction process needs to be halted for any reason such as the end of day has been reached, environmental conditions, or logistical issues, the construction halting procedure needs to be followed. Ensure the following:

- Bracing is installed on the final section of the tower.
- Any unstable or unsecured members are secured.
- Any loose materials are secured on site.
- Hoists and lifting materials are stored as per storage procedures.

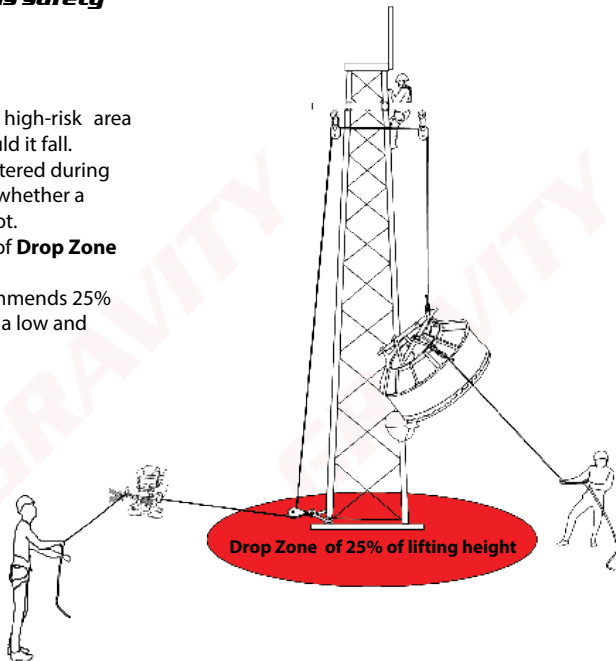
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1. Drop zone and hand tools safety

1. Drop Zones

- The drop zone refers to the high-risk area where equipment can land should it fall.
- The drop zone should not be entered during lifting operations, regardless of whether a person is wearing a helmet or not.
- Always ensure the correct ratio of **Drop Zone** when performing any lift.
- The industry best practice recommends 25% of the lifting height should it be a low and medium risk lift (4:1).



2. Hand tool safety

During the construction process a variety of hand tools will be used such as wrenches, torque wrenches, lever hoist etc. These tools pose a serious risk to the safety of the workers on site, should they be dropped.

- All hand tools must be secured with lanyards to the workers harness (or using a tool vest), up to 5 kg.
- All tools heavier than 5 kg must be lifted and lowered using a separate pulley system.
- All bolts, nuts and washers must be contained within a sealable tool pouch.
- When a bolt or nut is dropped to the ground the bolt may become brittle due to the high impact forces (shock load) that change the metal's physical characteristics. Never re-use dropped bolts or nuts.



3. Tool list

	Tool box		Podger spanner and or Pry bar
	Spirit level		Dumpy level, Staff and tripod
	Egyptian water level hose		Tape measure and Vernier
	Mallet hammer		Half round file
	Torque wrench		Cold galvanising spray

J. Tightening methods

1. Hand tighten and snug fit.

Hand tighten

The term “hand tighten” refers to tightening a fastener or component using only your hand, without the use of any tools or excessive force. It typically means tightening the fastener until it is securely in place, but without applying additional torque beyond what can be achieved by hand.

Hand tighten could allow for movement or play between connections, so as to assist in the assembly process.



Snug fit

A snug fit refers to a close and secure fit between two objects or components. It means that there is no extra space or gaps between them when they are joined together or placed in close proximity. A snug fit ensures that the objects fit tightly and securely without any unnecessary movement or play.



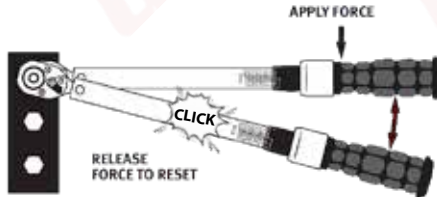
2. Torquing specifications

The torque specifications/torque values when tightening fasteners should be supplied with the TD.

Torque wrench:

Using a torque wrench properly is crucial for achieving accurate and consistent torque measurements. Here's a step-by-step guide on how to use a torque wrench:

1. Select the right torque wrench: Ensure you have the correct type and size of torque wrench for the job. Consider the torque range required for the specific task and choose a wrench that can accommodate that range and ensure that the wrench is calibrated.
2. Adjust the torque setting: Set the desired torque value on the torque wrench. Most torque wrenches have a handle or knob that allows you to adjust the torque setting. Refer to the manufacturer's instructions for the specific method of adjusting the torque setting.
3. Prepare the fastener: Ensure that the fastener you are working with is clean and properly lubricated if necessary. A clean and well-prepared fastener helps achieve accurate torque readings.
4. Attach the socket or appropriate tool: Connect the appropriate socket or tool to the square drive of the torque wrench. Make sure it securely attaches to the wrench to prevent any slippage during use.
5. Apply the torque: Place the socket or tool onto the fastener you wish to tighten. Apply steady and even force on the wrench in the direction specified for tightening (usually clockwise). Continue to apply force until you feel or hear the torque wrench click or release.



6. Release the tension: Once the torque wrench clicks or releases, stop applying force immediately. This indicates that the desired torque value has been reached. Do not over-torque the fastener beyond the specified torque value and do not reuse torqued bolts.
7. Reset or store the torque wrench: After using the torque wrench, it's important to reset the torque setting to zero or store it properly, depending on the wrench model. Refer to the manufacturer's instructions for the specific procedure.

Remember to always follow the torque specifications provided by the equipment manufacturer or relevant guidelines for the specific application you're working on. Using a torque wrench helps ensure proper tightening of fasteners and helps prevent over-tightening or under-tightening, which can result in mechanical failures or other issues.

Class 8.8 hot dip galvanizing bolt torque chart

Hot dip galvanized friction coefficient is about 0.20 to 0.35

Bolt size	Torque in Nm with f/c 0.20
M10	68
M12	117
M16	291
M24	1011

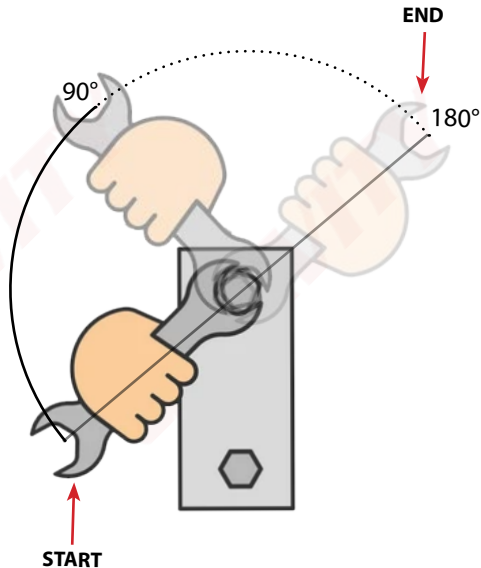
If the torque specifications/torque values are not supplied on the TD, one can refer to either the turn-of-the-nut method or refer to the torque chart, specific to the type of bolts being used.

Bolt tightening: turn-of-the-nut method

As per the mechanical engineering design handbook:

- Screws, fasteners, and design of non-permanent joints
- The "turn-of-the-nut method" indicated for heavy hexagonal structural bolts, states that the nut should be turned a minimum of 180 degrees from the snug-tight condition under optimum conditions.

The "turn-of-nut" is a commonly used and an accepted method, used extensively the tower industry. It is easy to use and results in a uniform bolt pretension. If a torque is specified to tighten the bolted connections, the bolt pretension is affected by the friction on the thread between the nut and the bolt as well as the friction between the connecting surfaces of the nut and the washer. The coefficients of friction at these two positions are affected by the type of surface (e.g. galvanized), the surface finish as well as the presence and type of lubricants. Since these parameters may differ from fastener to fastener, the torque method might not necessarily result in a uniform bolt pretension.



Notes:

3. Requirements of dismantling the tower.

Sometimes a tower needs to be dismantled due to the structure being too old or they need to upgrade the tower, or they are not receiving enough traffic on this tower to justify having it.

Dismantling of telecommunication towers is a different process to building a tower.

One of the main reasons is that the towers are often rusted, and when loosening the members onto the rigging line it could drop and shock load the system. If the tension in the rope is too high, the member could shoot up as it dislodges. This could lead to equipment damage or serious personal injury or even death. One cannot assume that if you can build a tower, you can dismantle it as well.

1. Ensure that the lifting rope and lever hoist is secured to the Gin pole before lifting it into place.
2. Install the Gin pole on the inside of the tower and jump it all the way to the top and secure it in the correct orientation and support it with the lever hoists.
3. Once the Gin pole has been secured, one can start to connect the lifting rope to the member that needs to be removed.
4. Secure 2 guide ropes to the member that will be removed and place a little bit of tension on the lifting system.
5. Start to loosen the bolts of the member that has slight tension.
6. It is crucial that when tension is placed on the member that needs to be lifted, it is just the right amount of tension. If the amount of tension is too little the member will drop and shock load the lifting system. If too much tension is placed on the lifting system the member will jump and can hit someone or damage the tower.
7. Once the member is loose it can be lowered to the ground and guided with the guide rope. Keep in mind that members loosened might be under tension.

Repeat this process by ensuring that the Gin pole is always in the correct orientation and in line with the member that needs to be lifted to prevent tension towards the side. Lower the Gin pole to the next relevant level and continue until the last leg remains. Then move the Gin pole to the outside of the stub and secure it with the correct supports and lower the last leg.

K. Permanent lifeline system

1. Gravity Vertical System

The Gravity Vertical Systems (GVS) are fixed lifeline systems installed on a wide variety of towers and structures where vertical work at height needs to be performed. It is a very effective fall arrest system that allows the worker to move up and down the tower seamlessly. No intervention is needed to pass the intermediate sections. The glider moves up and down freely and the system will catch the worker should he fall. The GVS could have a removable or a built-in glider which can be stored in a glider enclosure on the tower so it doesn't need to be removed and can't be stolen.

GVS is a very common lifeline installed on telecommunication towers throughout Africa. It has to be installed by a certified installer and it gets quality assured through a special application. The GVS systems needs to be checked annually and certified to be safe for use for the next 12 months.

For installation or inspections of lifelines, contact Gravity directly for more details.



L. Working at height and rigging

Once the first legs have been installed, it is time to take the tower to the next level. Correct work at height, rigging, and lifting procedures is essential.

The following competencies are needed to ensure a safe work at height environment.

Site Supervisor:

- Training on how to implement the Fall Protection Plan (IFPP).
- Fall Arrest and Basic Rescue (FABR).
- Rope Rigging (RR).
- Mechanical lifting (ML) – If weights will exceed more than 100 kg.
- Tower Building (TB).





Riggers and Technicians (at least 3 riggers to be on site at a time):

- Fall Arrest and Basic Rescue (FABR).
- Rope Rigging (RR).
- Mechanical lifting (ML) – If weights will exceed more than 100 kg.

The site supervisor is to ensure that the following administrative documents are in place before work starts:

- A Fall Protection Plan has been developed by a competent safety manager and the site supervisor has been inducted.
- Lifting plan.
- Pre task risk assessment.
- Medical records.
- Training records.
- Gear inspection records and pre-use inspections are done.
- Rescue procedures are in place.

Risk assessment:

A risk assessment must be conducted by the site supervisor on site, daily, before work starts in order to ensure that all risks on site are controlled and mitigated.

If at any point the work site becomes unsafe or if a new risk arises it has to be revised, managed and communicated before work can continue.

Lifting plan:

A lifting plan is a document specifying how and by whom the lift will be completed. This must be completed by the site supervisor on site before work starts.

If at any point alterations have to be made to the lifting operation which were not indicated on the lifting plan, it must be revised, documented, communicated and signed by the supervisor and team before continuing with the lift.

How to climb safely

PPE requirements for basic climbing:

Helmet, harness, shock-absorber, and work belt.



In order to be able to climb the tower safely whilst constructing the rest of the sections, it is required to install a temporary vertical lifeline on the top hole of the top splice plate of every new leg member whilst it is still on the ground. It is important to not install the lifeline in the holes of the leg members as this will interfere with the installation of the bolts for the connector plates.

1. Work at height principles

Fall Factors

Fall Factor =

$$\frac{\text{Distance of fall}}{\text{Length of lanyard}}$$



Fall Factor 0:



Fall Factor 1:



Fall Factor 2:

2. Rope rigging

Rope rigging enables a technician to lift loads that weigh up to 100 kg by creating mechanical advantage systems while using pulleys, ropes, connectors, slings, and descenders.



Rigging Principles

- Always be in reach of your controlling device.
- Never stand directly below the load.
- Ensure the lifting route is clear of obstructions.
- Ensure that the slinging is done in such a way that the load is as secure in the air as it would be on the ground.
- The weight of the load should not exceed the SWL of the slinging gear.
- Always deviate from the bottom of the tower.

3. Mechanical lifting

Mechanical lifting enables a technician and his team to lift loads that weigh more than a 100 kg by creating mechanical advantage systems while using pulleys, ropes, connectors, slings, and mechanical machines to lift the loads.

The most common lifting machines used are a lever hoist, a tirfor (wire rope puller) and a Gravity Capstan hoist.



Gravity 500 Capstan
hoist
SWL (200 kg)



Gravity 1200
Capstan hoist
SWL (250 kg)

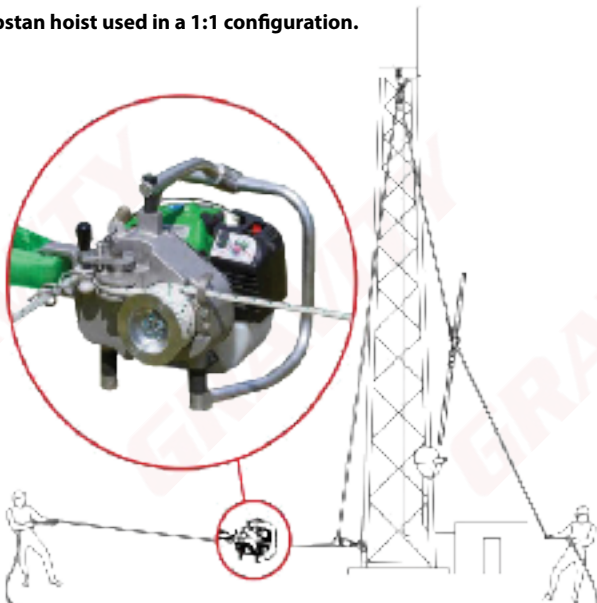


Tirfor



Lever hoist

Gravity 1200 Capstan hoist used in a 1:1 configuration.



4. Stop start process for Gravity Capstan hoist

Gravity 1200 Capstan hoist - 2 stroke.

- Fill the tank with the pre-mixed Petrol and 2 stroke oil to the indicated level of the tank.
- Fix the hoist in a working position.
- Bleed the primer about one or two times until the fuel is visible in the primer.
- Move the choke lever to the closed position.
- Firmly take hold of the starter handle and pull the starter once.
- Move the choke lever to the open position.
- Switch the kill switch ON.
- Push the decompression button down.
- Firmly take hold of the starter handle and pull the starter once.

- When starting a warm motor do not switch the choke lever to close.
- Switch the kill switch ON.
- Push the decompression button down.
- Firmly take hold of the starter handle and pull the starter once.

- To stop the motor, keep the rope slacked and ensure the throttle is set to idle.
- Allow the motor to cool by running it in idle for a few seconds.
- Turn the kill switch to the OFF position and the motor must stop.

- Push the decompression button down as an emergency kill switch to switch off the hoist.



Notes: _____

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M. Major and minor discrepancies

Minor discrepancies

- **Spacial awareness.**
- **Dropping of bolts.**

Major discrepancies

- **Spacial awareness multiple time under the load.**
- **Dropping of bolts more than 3 times.**
- **Dropping tower members.**
- **Climbing un-safely.**
- **Overloading lifting equipment.**
- **Using the Gin pole past the 25-degree angle.**
- **Climbing the Gin pole or using it as a fall arrest anchor.**
- **Anchoring on lifting equipment.**

N. DEFINITIONS

1. **Bill of material (BOM):** This is a part of the drawings of the tower which stipulates all the components of the tower, their weight, size, and part number.
2. **Bracing:** The horizontal and diagonal members of a tower, can be seen from the side view.
3. **Catastrophic failure:** A sudden and total failure of a structure from which recovery is impossible. At best, it will result in major financial loss, or at worst, it can also cost multiple lives.
4. **Connector plates:** These are components used to join horizontal, diagonal, and vertical members together.
5. **General assembly drawings:** The part of the technical drawings indicating an overview of the whole structure.
6. **Leg connectors:** These are components used to join two leg members of a tower together. On CHS (Circular Hollow Sections), these connectors are part of the leg and are called flanges. On Angle iron members these connectors are separate plates that get bolted to the legs.
7. **Planned bracing:** Internal horizontal members of a tower, this can be seen from the top view.
8. **Section drawing:** The part of the technical drawing indicating in more detail how a section looks, gets assembled, and often also contains a bill of material.
9. **Technical drawing (TD):** A detailed drawing of the tower, its components, and how it fits together.
10. **Vernier caliper:** A measuring device that is used for the measurement of linear dimensions. It is also used for the measurement of diameters of round objects with the help of the measuring jaws.



Congratulations on completing Gravity Training's Tower Erector course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.

